Equivalenze asintotiche

Def.

Siano a_n e b_n successioni, si dicono equivalenti

se $\frac{a_n}{b_n} \to 1$ $\left(a_n \sim b_n \right.$

$a_n = b_n (1 + o(1))$

$\left. b_n = a_n (1 + o(1)) \right)$

Per $a_n \to 0$

i) $\log(1+a_n) \sim a_n$

$$\frac{\log(1+a_n)}{a_n} \to$$

ii) $e^{a_n} - 1 \sim a_n$

$$\frac{e^{a_n}-1}{a_n} \to 1$$

$\forall \alpha \in \mathbb{R}$

iii) $(1+a_n)^\alpha \sim 1 + \alpha a_n$

$$(1+a_n)^\alpha - 1 \sim \alpha a_n$$

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Per $|b_n| \to \infty$

i) $\left(1 + \frac{c}{b_n}\right)^{b_n} \sim e^c$

$c=1$ $\left(1+\frac{1}{b_n}\right)^{b_n} \sim e$

$c=-1$ $\left(1-\frac{1}{b_n}\right)^{b_n} = \left(1+\frac{(-1)}{b_n}\right)$

$\sim e^{-1} = \frac{1}{e}$

Funzioni trigonometriche

Per $a_n \to 0$

i) $\sin(a_n) \sim a_n$

$$\frac{\sin a_n}{a_n} \to 1$$

ii) $1 - \cos(a_n) \sim \frac{1}{2} a_n^2$

iii) $\operatorname{tg}(a_n) \sim a_n$

Ese 1

$$\lim_{n\to\infty} \frac{\sin\left(\frac{3}{n}\right)}{\frac{3}{n}} \sim \frac{\frac{3}{n}}{\frac{3}{n}} = 1$$

$a_n = \frac{3}{n}$ per $n \to \infty$

$a_n = \frac{3}{n} \to 0$ ✓

Ese 2

$$\lim_{n\to\infty} n \cdot \sin\left(\frac{2}{n}\right) \sim \cancel{n} \cdot \frac{2}{\cancel{n}} = 2$$

$a_n = \frac{2}{n}$

per $n \to \infty$

$a_n \to 0$ ✓

Ese 3

$$\lim_{n\to\infty} n^2 \left(1 - \cos\left(\frac{3}{n}\right)\right) \sim n^2 \cdot \frac{1}{2} \cdot \left(\frac{3}{n}\right)^2 =$$

$$= \cancel{n^2} \cdot \frac{1}{2} \cdot \frac{9}{\cancel{n^2}} = \frac{9}{2}$$

Ese 4

$$\lim_{n\to\infty} n\tan\left(\frac{4}{n}\right) \sim n\cdot\frac{4}{n} = 4$$

Ese 5

$$\lim_{n\to\infty} \frac{\sin\left(\frac{5}{n}\right)}{\frac{1}{n^2}-\frac{2}{n}} = \lim_{n\to\infty} \frac{\sin\left(\frac{5}{n}\right)}{\frac{1}{n}\left(\frac{1}{n}-2\right)}$$

$$\sim \frac{\frac{5}{n}}{\frac{1}{n}\left(\frac{1}{n}-2\right)} = \frac{5}{-2} = -\frac{5}{2}$$

($\frac{1}{n}\to 0$)

Esercizio 6

$$\lim_{n\to\infty} \frac{\tan\left(\frac{1}{n}\right) - \sin\left(\frac{1}{n}\right)}{\frac{1}{n^3}} = n^3\left[\tan\left(\frac{1}{n}\right) - \sin\left(\frac{1}{n}\right)\right]$$

$$\tan x = \frac{\sin x}{\cos x}$$

NUM.

$$\tan\left(\frac{1}{n}\right) - \sin\left(\frac{1}{n}\right) = \frac{\sin\left(\frac{1}{n}\right)}{\cos\left(\frac{1}{n}\right)} - \sin\left(\frac{1}{n}\right) =$$

$$= \sin\left(\frac{1}{n}\right)\left[\frac{1}{\cos\left(\frac{1}{n}\right)} - 1\right] =$$

$$= \sin\left(\frac{1}{n}\right)\left[\frac{1-\cos\left(\frac{1}{n}\right)}{\cos\left(\frac{1}{n}\right)}\right] = \sin\left(\frac{1}{n}\right)\cdot\left(1-\cos\left(\frac{1}{n}\right)\right)\cdot\frac{1}{\cos\left(\frac{1}{n}\right)}$$

$$\lim_{n\to\infty} n^3\cdot\sin\left(\frac{1}{n}\right)\cdot\left(1-\cos\left(\frac{1}{n}\right)\right)\cdot\frac{1}{\cos\left(\frac{1}{n}\right)} =$$

$$\lim_{n\to\infty} \underbrace{n\cdot\sin\left(\frac{1}{n}\right)}_{\frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}}\to 1}\cdot\underbrace{n^2\cdot\left(1-\cos\left(\frac{1}{n}\right)\right)}_{1-\cos\left(\frac{1}{n}\right)\to\frac{1}{2}\left(\frac{1}{n}\right)^2}\cdot\underbrace{\frac{1}{\cos\left(\frac{1}{n}\right)}}_{\to 1} \sim$$

$$\sim 1\cdot\frac{1}{2}\cdot 1 = \frac{1}{2}$$

Ese 7/

$$\lim_{n\to\infty} \frac{\cos\left(\frac{1}{n^3}\right) - e^{\frac{2}{n^3}}}{\tan\left(\frac{1}{n^3}\right)} =$$

$1-\cos(a_n) \sim \frac{1}{2}a_n^2$

$e^{a_n} - 1 \sim a_n$

$$= \lim_{n\to\infty} \frac{\cos\left(\frac{1}{n^3}\right) - 1 + 1 - e^{\frac{2}{n^3}}}{\tan\left(\frac{1}{n^3}\right)} =$$

$$= \lim_{n\to\infty} -\frac{1-\cos\left(\frac{1}{n^3}\right) + e^{\frac{2}{n^3}} - 1}{\tan\left(\frac{1}{n^3}\right)} \sim -\frac{\frac{1}{2}\cdot\left(\frac{1}{n^3}\right)^2 + \frac{2}{n^3}}{\frac{1}{n^3}} =$$

$\to 0$

$$= (n^3)\cdot(-1)\cdot\left(\frac{1}{2n^6} + \frac{2}{n^3}\right) = (n^3)(-1)\cdot\frac{1}{n^3}\left(\frac{1}{2n^3}+2\right)$$

$= (-1)(2) = -2$

Esercizio 8

$$\lim_{n\to\infty} \frac{n^2\cdot\left[e^{\sin\left(\frac{a}{n}\right)}-1\right]}{\sqrt{3+n^2}+\sqrt{n^2+1}} \sim$$

$a\in\mathbb{R}$, $a\neq 0$

$\sin(a_n)\sim a_n$

$e^{a_n}-1\sim a_n$

$$\sim \frac{n^2\left[e^{\left(\frac{a}{n}\right)}-1\right]}{\sqrt{3+n^2}+\sqrt{n^2+1}} \sim n^2\cdot\frac{a}{n} = \frac{n\cdot a}{\sqrt{3+n^2}+\sqrt{n^2+1}} =$$

$$= \frac{n\cdot a}{\sqrt{n^2\left(\frac{3}{n^2}+1\right)}+\sqrt{n^2\left(1+\frac{1}{n^2}\right)}} = \frac{\cancel{n}\cdot a}{\cancel{n}\left(\sqrt{\frac{3}{n^2}+1}+\sqrt{1+\frac{1}{n^2}}\right)} = \frac{a}{2}$$

$\frac{3}{n^2}\to 0$, $\frac{1}{n^2}\to 0$

Esercizio

$$\lim_{n\to\infty} \frac{e^{\sin\left(\frac{a}{n}\right)} - 1}{\sqrt{3+n^2} - \sqrt{n^2+1}} \sim$$

$a \in \mathbb{R}$, $a \neq 0$

$$\left(\sqrt{3+n^2} - \sqrt{n^2+1}\right) \cdot \frac{\left(\sqrt{3+n^2} + \sqrt{n^2+1}\right)}{\sqrt{3+n^2} + \sqrt{n^2+1}} =$$

$$= \frac{3 + n^2 - n^2 - 1}{\sqrt{3+n^2} + \sqrt{n^2+1}} = \frac{2}{n\sqrt{\frac{3}{n^2}+1} + n\sqrt{1+\frac{1}{n^2}}} = \frac{2}{n\left(\sqrt{\frac{3}{n^2}+1} + \sqrt{1+\frac{1}{n^2}}\right)}$$

$$\sim \frac{a}{n} \cdot \frac{n\left[\sqrt{\frac{3}{n^2}+1} + \sqrt{\frac{1}{n^2}+1}\right]}{2} = \frac{a \cdot 2}{2} = a$$

($\frac{3}{n^2} \to 0$, $\frac{1}{n^2} \to 0$)

Ese 10

$$\lim_{n\to\infty} \sqrt{1+\sin\left(\frac{2}{n}\right)} = \lim_{n\to\infty} \left[1+\sin\left(\frac{2}{n}\right)\right]^{\frac{1}{2}}$$

$$\sim \left(1+\frac{2}{n}\right)^{\frac{1}{2}} \sim 1+\frac{1}{2}\cdot\frac{2}{n} = 1+\frac{1}{n} = 1$$

Ese 11

$$\lim_{n\to\infty} n\left[\log\left(5e^{\frac{1}{n}} - 4 - \frac{1}{n}\right) - \sin\left(\frac{3}{n}\right)\right]$$

$$= \lim_{n\to\infty} n\left[\log\left(5\,e^{\frac{1}{n}} - 4 - 1 + 1 - \frac{1}{n}\right) - \sin\left(\frac{3}{n}\right)\right]$$

$$= \lim_{n\to\infty} n\left[\log\left(5\left(e^{\frac{1}{n}} - 1\right) + 1 - \frac{1}{n}\right) - \sin\left(\frac{3}{n}\right)\right]$$

$$\sim n\left[\log\left(5\cdot\frac{1}{n} + 1 - \frac{1}{n}\right) - \frac{3}{n}\right] = n\cdot\left[\log\left(\frac{4}{n} + 1\right) - \right.$$

$$\sim \quad n \cdot \left[\frac{4}{n} - \frac{3}{n} \right] = n \cdot \frac{1}{n} = 1$$

$\frac{3}{n}\Big] =$